

Worksheet - Insect Research

Note: For each insect, open the labelling task for Image 2 in the simulation. Activate labelling mode, assign the terms correctly, export the screenshot and insert it into the box provided on the following pages.

Introduction

Insect legs show how structure fits function. All legs have the same basic sections: coxa, trochanter, femur, tibia and tarsus. Depending on habitat and task, these sections are modified for running, digging, swimming, collecting, catching or jumping.

Task 1: Ground beetle

Examine the ground beetle's standard leg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Ground beetle
Scientific name	Carabus violaceus

Matching leg type: _____

Insert screenshot of Image 2 here

Task: Explain how this leg's structure fits its function.

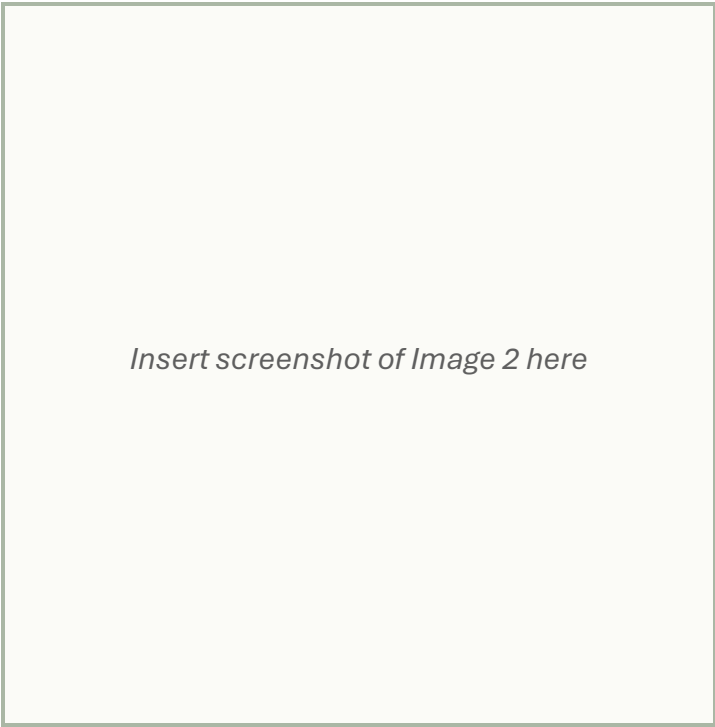
Task: Striking features in Image 3 ("running leg"):

Task 2: Mole cricket

Examine the mole cricket's foreleg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Mole cricket
Scientific name	Gryllotalpa gryllotalpa

Matching leg type: _____



Task: Explain how this leg's structure fits its function.

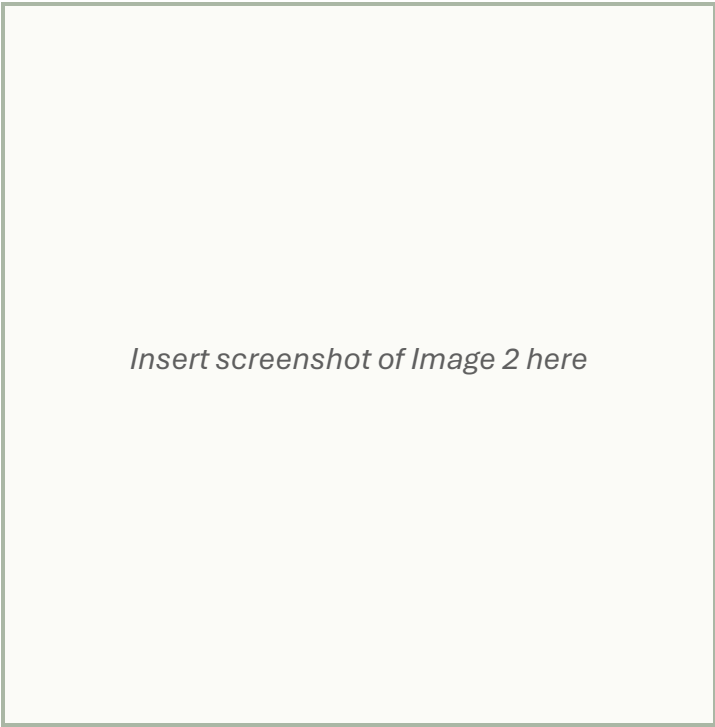
Task: Striking features in Image 3:

Task 3: Diving beetle

Examine the diving beetle's foreleg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Diving beetle
Scientific name	Dytiscus marginalis

Matching leg type: _____



Task: Explain how this leg's structure fits its function.

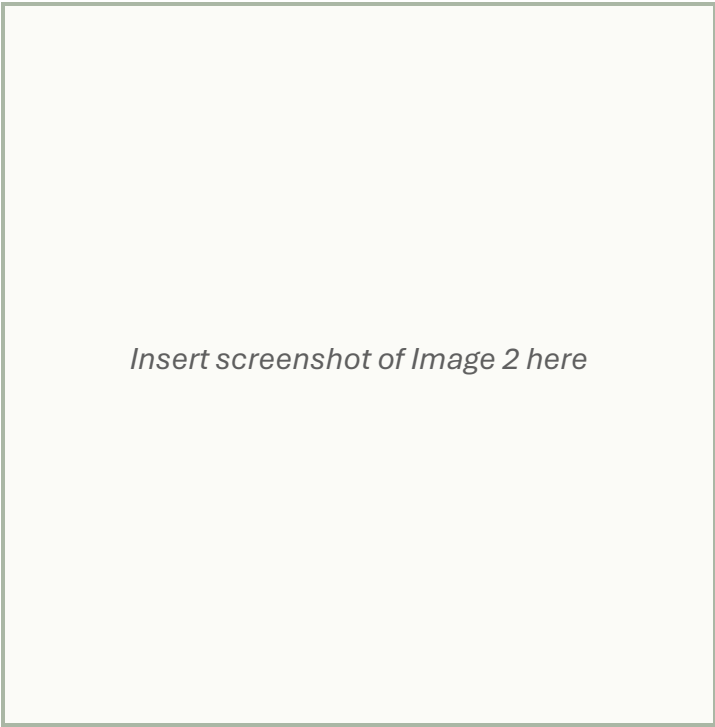
Task: Striking features in Image 3:

Task 4: Bee

Examine the bee's foreleg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Bee
Scientific name	Apis mellifera

Matching leg type: _____



Task: Explain how this leg's structure fits its function.

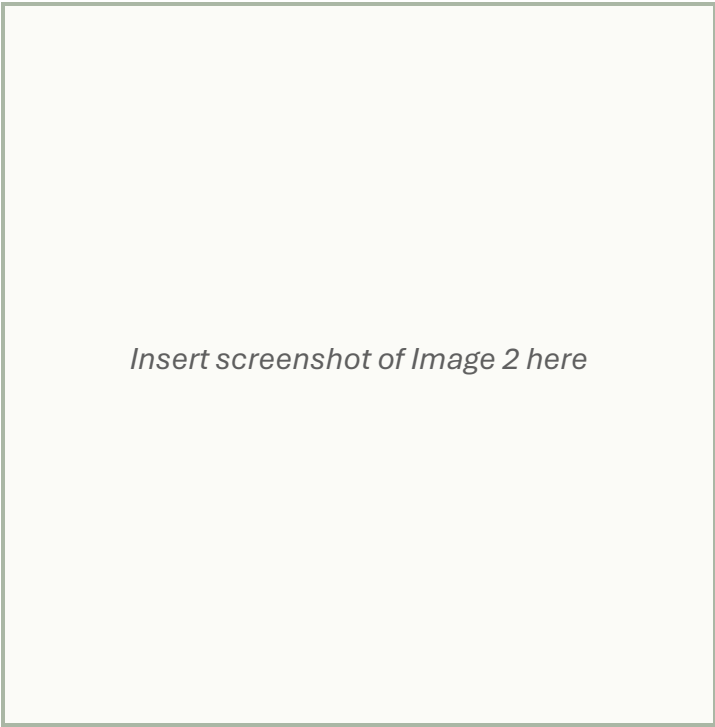
Task: Striking features in Image 3:

Task 5: Praying mantis

Examine the praying mantis's foreleg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Praying mantis
Scientific name	Mantis religiosa

Matching leg type: _____



Task: Explain how this leg's structure fits its function.

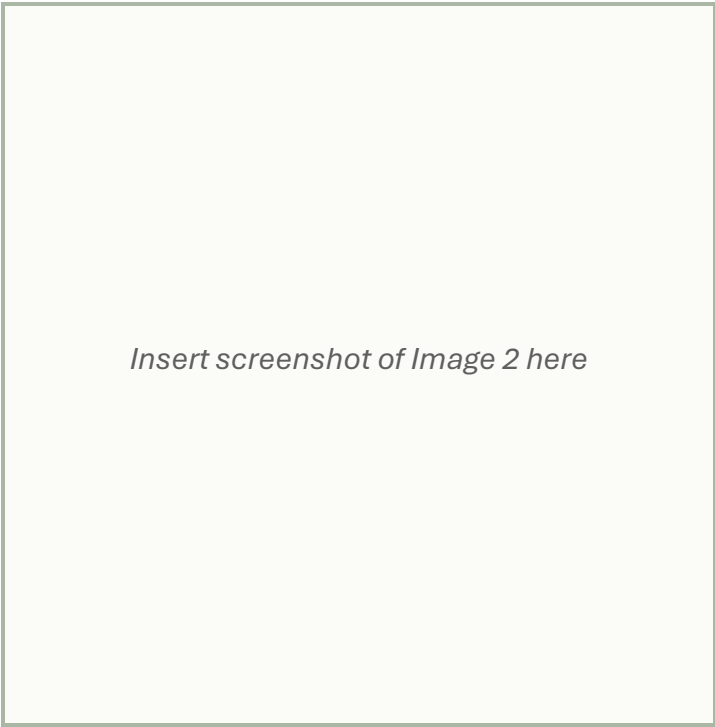
Task: Striking features in Image 3:

Task 6: Grasshopper

Examine the grasshopper's foreleg. Use Image 1 and Image 3 to investigate its structure.

Associated insect	Grasshopper
Scientific name	Chorthippus sp.

Matching leg type: _____



Task: Explain how this leg's structure fits its function.

Task: Striking features in Image 3:

Task 7: Overview and summary

Summarise your results. Compare the leg types and draw a justified conclusion.

Insect	Leg type	Striking features
Ground beetle		
Mole cricket		
Diving beetle		
Bee		
Praying mantis		
Grasshopper		

Task: Explain how insect legs show the relationship between structure and function.

Task: Create a hypothesis about the legs of flying insects, such as mosquitoes, butterflies or dragonflies. Justify it using structure and function.
